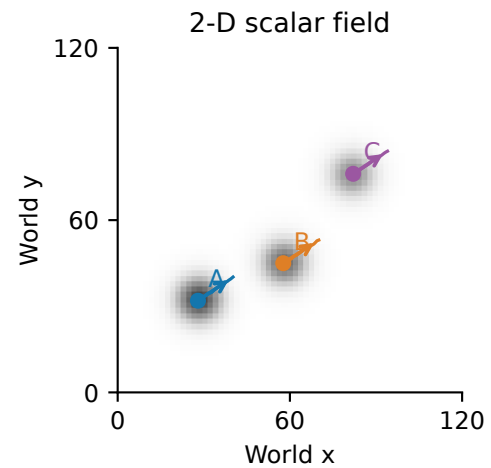
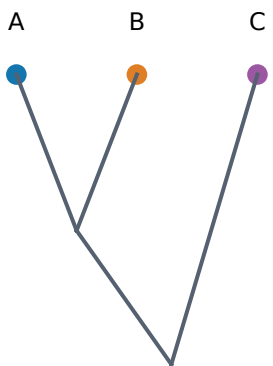


Dual-Reference RA-MTM: two complementary views, one feature identity



Augmented merge tree
(schematic hierarchy)



Leaf supports

$$C_i = (C_i^x, C_i^y)$$

$$A_i$$

Shared identity
and hierarchy

Fixed X projection

$$q_i^x = a_x^T C_i$$

Hierarchy-constrained RA

$$w_i = cA_i$$

$$|x_i - q_i| \leq \tau^* + \Delta$$

2-D geometry +
motion residual

Fixed Y projection

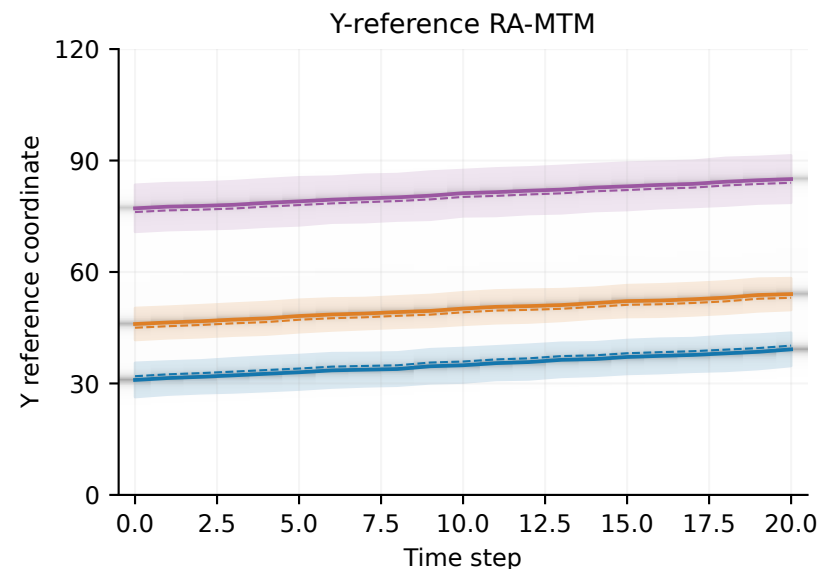
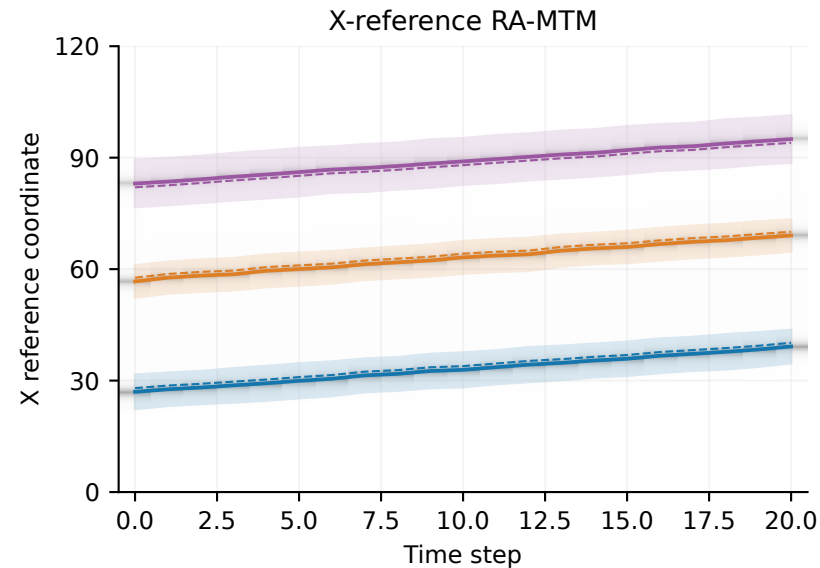
$$q_i^y = a_y^T C_i$$

Hierarchy-constrained RA

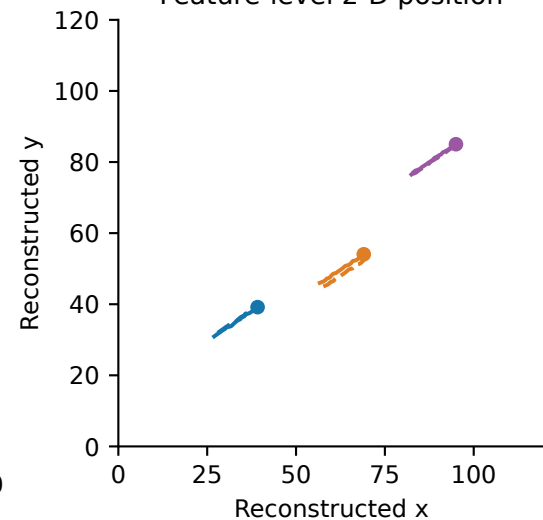
$$w_i = cA_i$$

$$|x_i - q_i| \leq \tau^* + \Delta$$

2-D geometry +
motion residual



Feature-level 2-D position



$$\hat{C}_i = (x_i^x, x_i^y)$$

Matched identity
across both views

2-D motion from
position differences

No full scalar-field inverse